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Assignment - 3

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S046
MBA TECH DS
J2

Q.1 $P(T) = 0.5$
$P(B|T) = 0.6$, $P(R) = 0.3$, $P(P) = 0.6$
$P(B|R) = 0.3$, $P(B|P) = 0.1$

Using Bayes Theorem:

$$P(R|B) = \frac{P(R)\,P(B|R)}{P(T)\,P(B|T) + P(R)\,P(B|R) + P(P)\,P(B|P)}$$

$$= \frac{0.3(0.3)}{0.5(0.6) + 0.3(0.3) + 0.6(0.1)}$$

$$= \frac{0.09}{0.42} = \frac{3}{14} \approx 0.214$$

(Ans)

Q.2 $P(B) = 0.8$
$P(L) = 0.2$
$P(F|B) = 0.30$, $P(F|L) = 0.95$

$$P(B|F) = \frac{0.8(0.30)}{0.8(0.3) + 0.2(0.95)}$$

$$= \frac{0.24}{0.24 + 0.19} = \frac{0.24}{0.43} \approx 0.558$$

For LPG:

$$P(L|F) = \frac{0.19}{0.43} \approx 0.442 \text{ or } 44\%$$

(Ans)

$\therefore$ Short circuit is ~~the~~ more probable ~~than~~ cause.

Q.3 (i) $P(HHH) = \frac{1}{5} + \frac{1}{10} = \frac{3}{10} = 0.3$

(ii) Probability it was the two - headed coin

$$P(2H \mid HHH) = \frac{\frac{1}{5}(1)}{\frac{1}{5}(1) + \frac{4}{5}\left(\frac{1}{8}\right)}$$

$$= \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{10}} = \frac{2}{3} \approx 0.67$$

or 67% (Ans)

Q.4 Three jewellery boxes

Box 1: gold, gold

Box 2: silver, silver

Box 3: gold, silver

Possible silver drawers:

Box 2 has 2 silver drawers

Box 3 has 1 silver drawers

$\therefore$ P(other drawer is gold | silver observed) $= \frac{1}{3}$

~~or~~ 0.33

33%

(Ans)

$\frac{R}{13/18}$

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